AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 31 · STUDENT LESSON

Subgroup Performance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the slice. Count its cases. Keep the total honest.

Disaggregate error and recall without hiding missing coverage or claiming causes from observed gaps.

Keep one habit: Complete main counts → pictured denominator change → full lighting/type/camera audit → independent new groups.

MAIN LESSON · SECTION 1

Which condition does the overall result conceal?

> Keep subgroup definitions,numerators and denominators beside every comparison.

Learning goals

I can calculate **subgroup** results with sample sizes.

I can distinguish missing **evidence** from equal performance.

I can report differences without assigning causes from a table.

Recall pooled-rate limits

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.
- What can a pooled error rate hide?

End goal

By the end, I can explain **subgroup** performance using a traced example and a stated limitation.

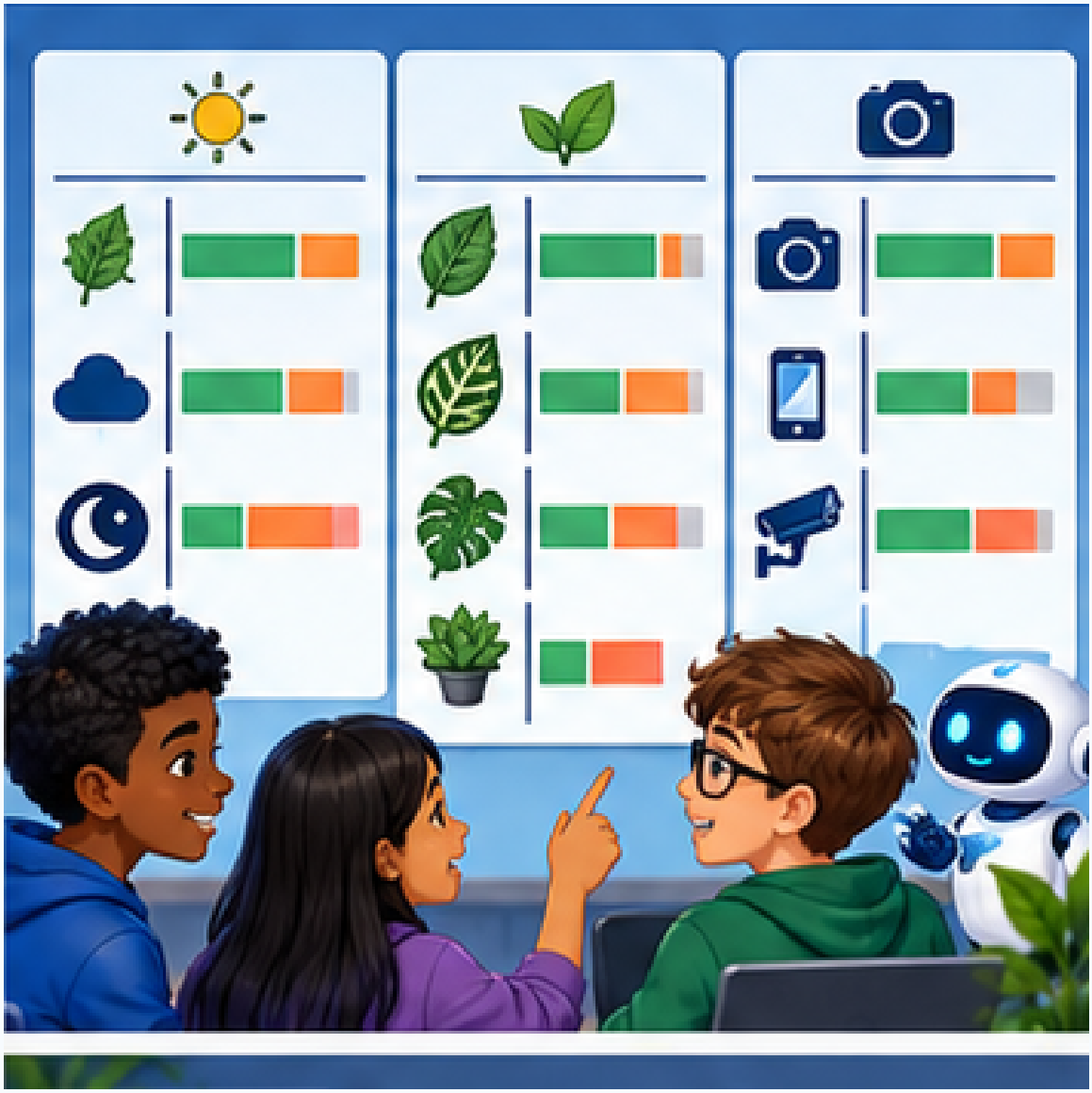
▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.



LEARNING TARGET Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Declare meaningful object conditions before inspecting results

> This lesson uses fictional lighting,toy plant types and cameras,not personal identities.

Disaggregation explanation and fictional scope

A single overall score can hide different experiences. Define relevant groups before evaluation and report counts as well as rates. This classroom exercise uses fictional sensor conditions, not personal identities.

Real evaluations involving people require appropriate permissions, privacy and contextual expertise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

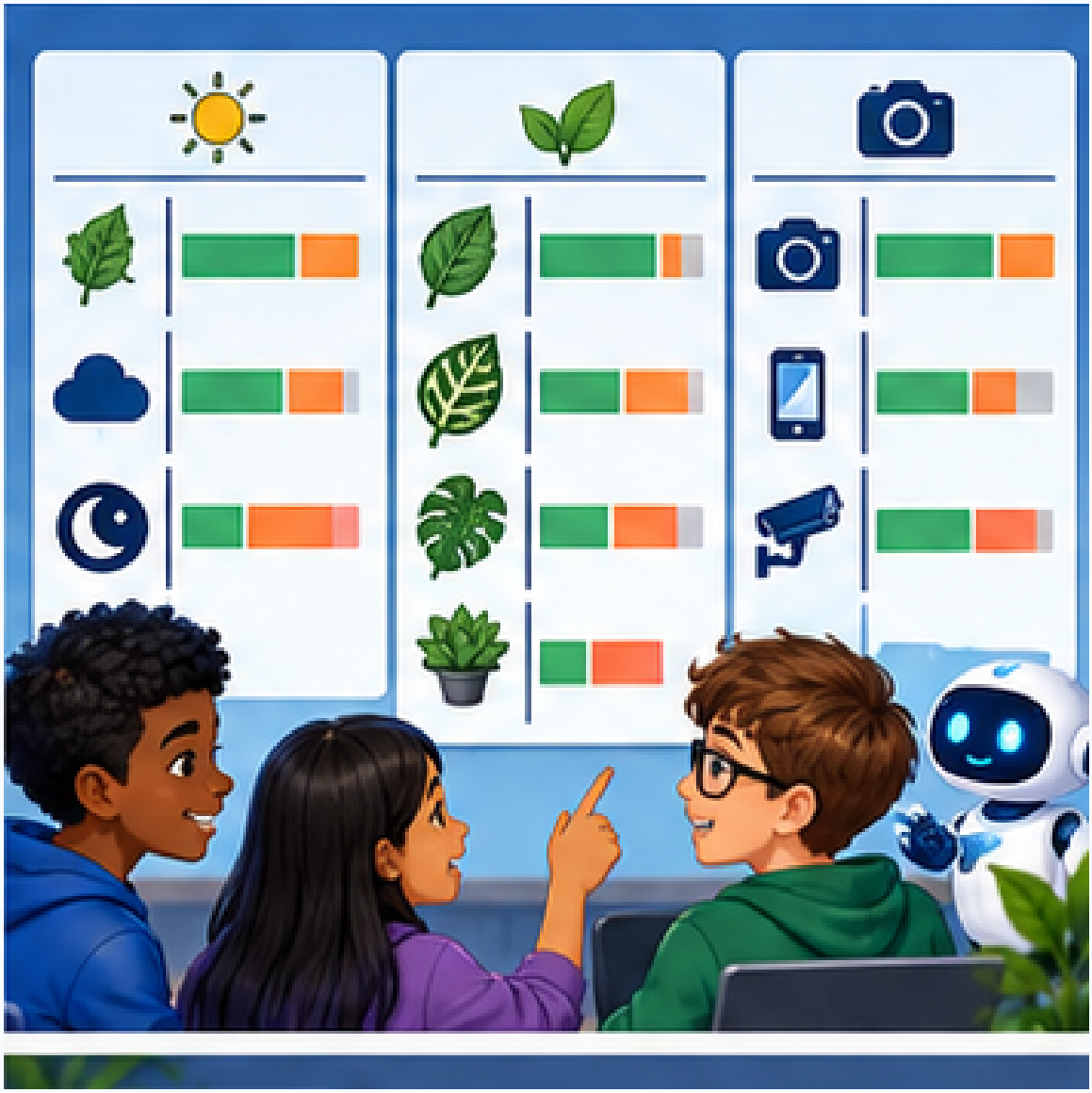
Earlier weeks supplied the model-building **process**; this week examines **subgroup** evaluation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **subgroup** performance to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Trace the pooled rate and the small-slice sensitivity

> The larger sample can dominate a headline without eliminating the smaller group’s evidence gap.

Complete main count/rate table

Main sensor/camera slice	Observed errors	Evaluated count	Error rate / accuracy
A	2	20	10% /90%
B	3	5	60% /40%
Pooled disjoint A/Bcases	5	25	20% /80%
Observed error-rate difference	Bminus A	Same error definition	50 percentage points

Sensor group A has 2 errors in 20 examples and group B has 3 in 5. Error rates are 10 and 60 percent; pooled error is 5/25=20 percent. The group difference is 50 percentage points.

B has only five examples, so its rate is sensitive to each individual result.

Explain weighting,sensitivity and supplied-record scope

The pooled error is a count-weighted result, not the unweighted average of10%and 60%. With only 5 cases, one changed Bresult moves its error rate by1/5=20 percentage points. The complete Slice Audit below supplies one synthetic case-level realization matching these aggregate camera counts; it adds explicit lighting, type and toy reference labels for arithmetic.

It does not reveal a causal camera effect or establish real biological performance.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

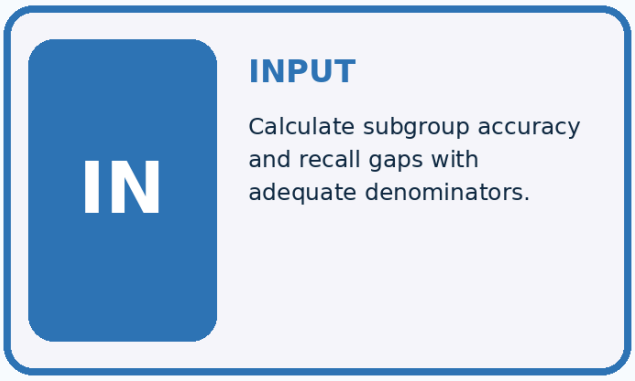
Keep the distinction between an observed result and a proposed next step.

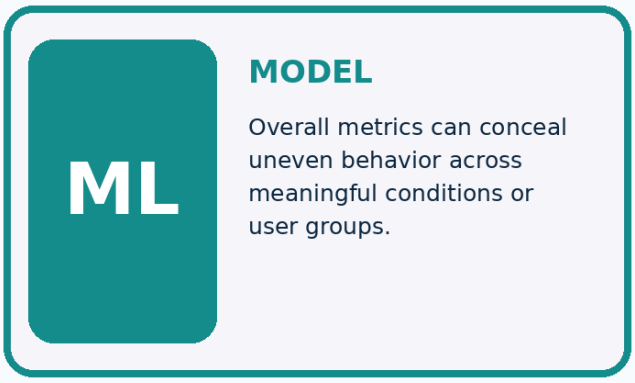
▼ Earlier wording, alternatives and picture transcripts

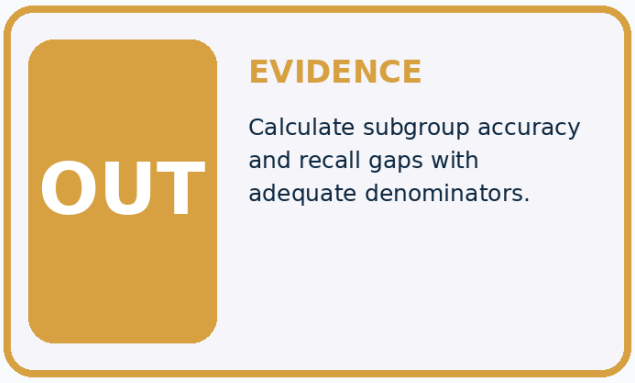
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Calculate **subgroup** accuracy and recall gaps with adequate **denominator**s.

MODEL IDEA Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Daylight: 18/20 correct = 90%; evening: 10/20 = 50%. Accuracy gap = 90% – 50% = 40 percentage points.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

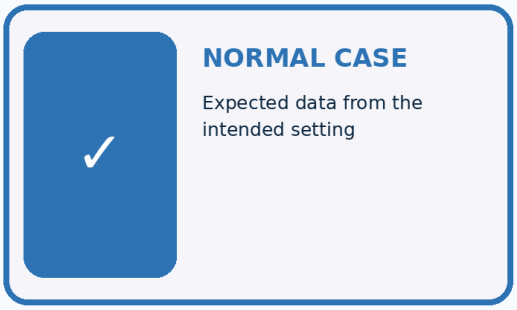
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

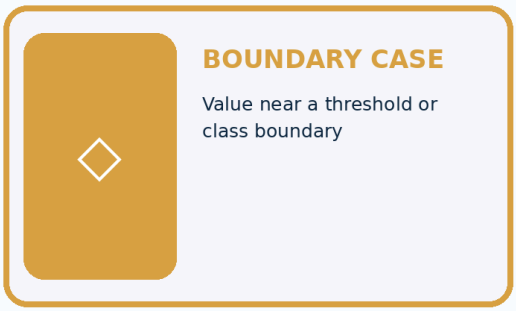
Compare a total score with the conditions behind it

> A rate difference needs its sample counts and a source-quality investigation.

Four evaluation conditions



NORMAL Expected data and expected behavior.

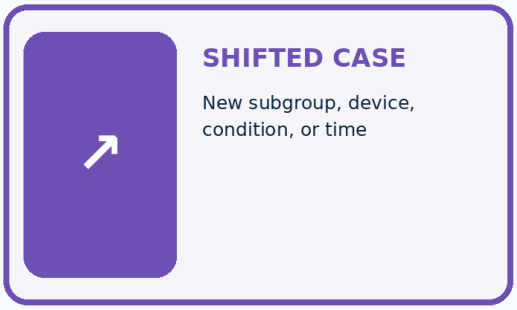


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, **subgroup**, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Shade gains5 new cases with no errors.

▼ **Check after predicting**

Prediction: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

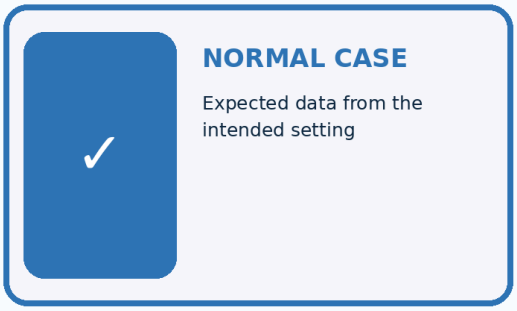
Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

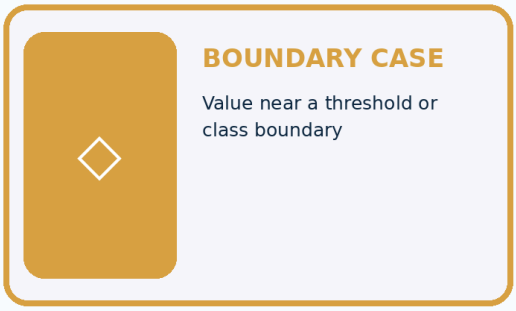
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

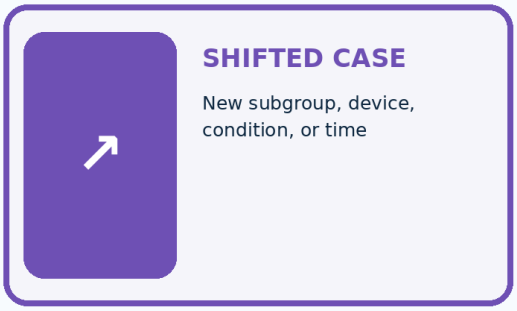
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, **subgroup**, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the added cases and unchanged errors beside the picture

> A denominator change is different from a repaired classifier.

Complete pictured facts and change scope

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

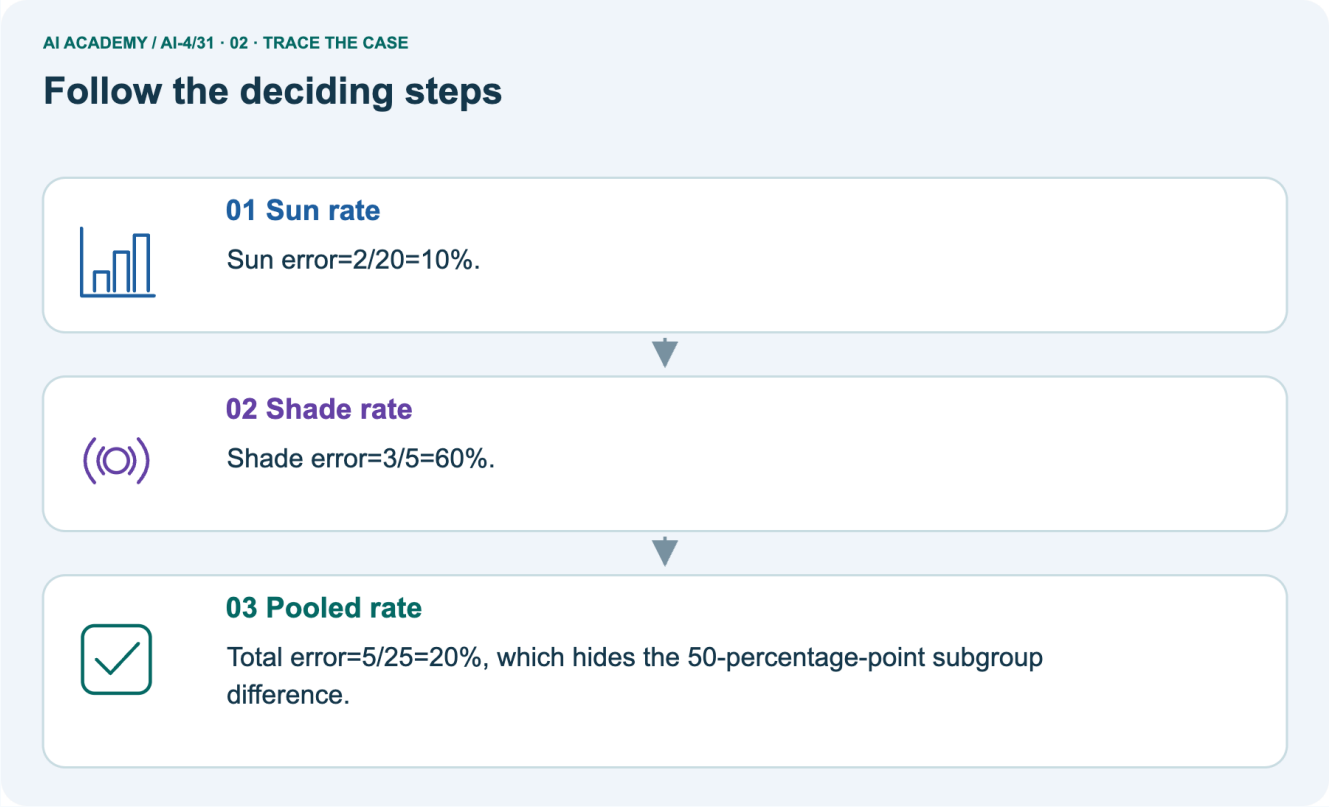
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Picture Sun2/20 errors and Shade3/5 match the main 10%/60%arithmetic; pooled 5/25=20%, gap 50 percentage points. Adding five new Shadecases with no errors changes Shade to3/10=30%and pooled to5/30≈16.7%, without removing any original error. The model was

not improved merely by changing the evaluation sample. New per-class labels are not supplied for those additions, so do not invent revised recall.

Correcting one of the original five Shadeoutputs instead would give 2/5=40%; record a reviewed output or model-version change separately.

Trace Sun,Shade and pooled rates



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

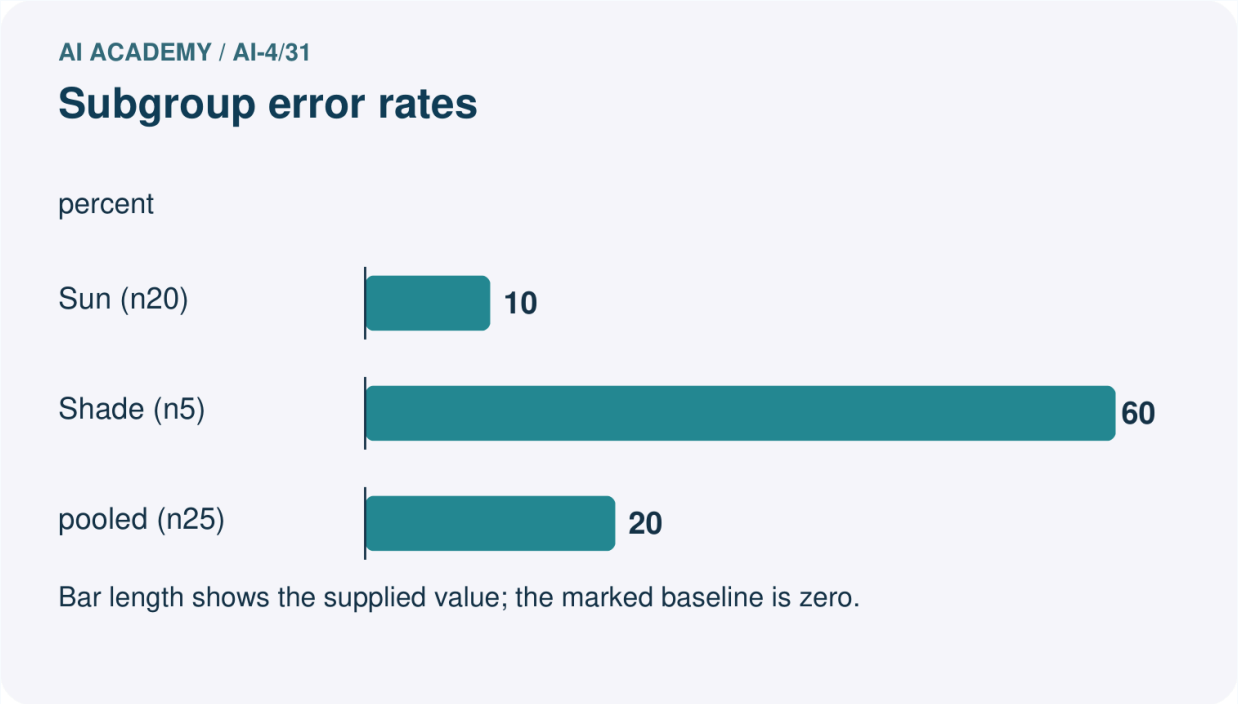
STEP	WHAT HAPPENS AND WHY
1. Sun rate	Sun error=2/20=10%.
2. Shade rate	Shade error=3/5=60%.
3. Pooled rate	Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original subgroup comparison graphic

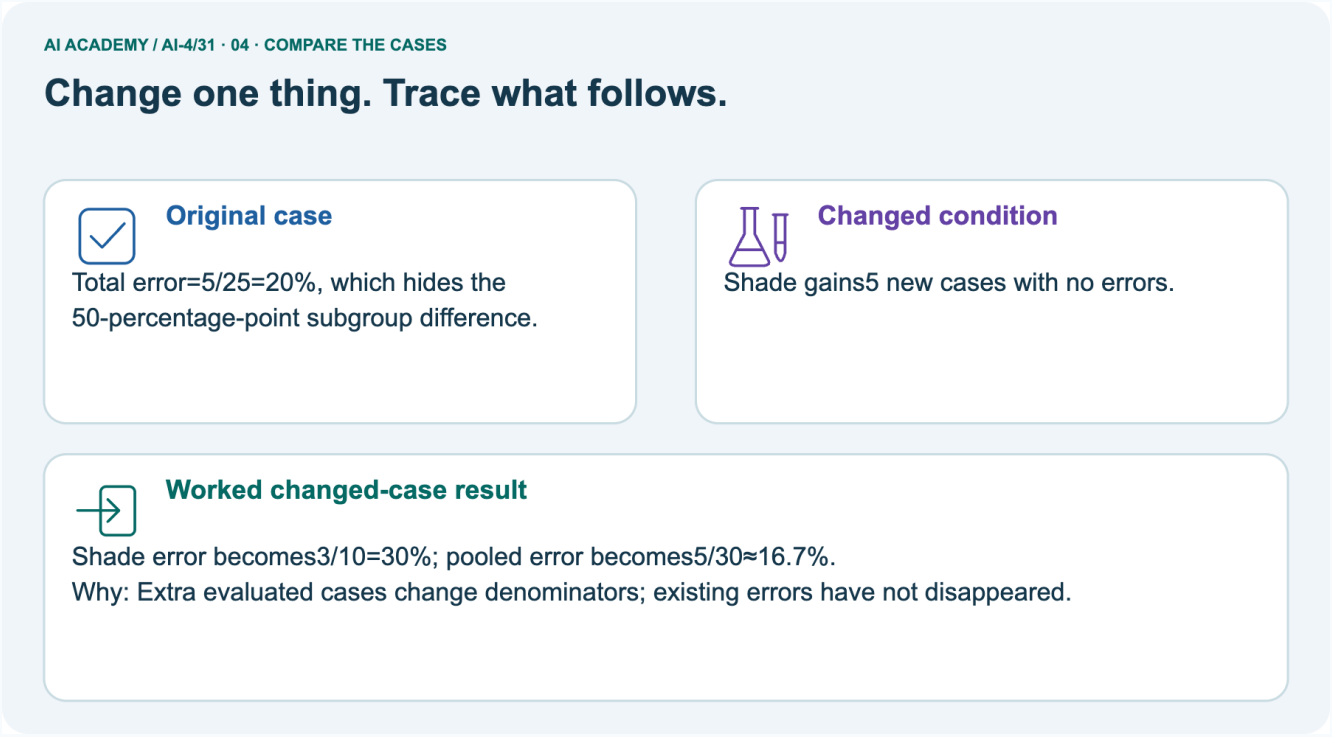


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict adding five error-free Shadecases

- Change one thing: Shade gains5 new cases with no errors. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain sampling and causal limits

- Small **subgroup** samples have high variability. An observed gap does not by itself establish discrimination or a causal explanation.

- Read the labelled picture: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to one-error sensitivity

Subgroup evaluation at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Group A	2 errors of 20	10 percent
Group B	3 errors of 5	60 percent
Pooled	5 errors of 25	20 percent

Changing one B error to a correct result changes its rate from 60 to 40 percent. Small samples can be unstable without making the observed errors unimportant. A missing group has no measured rate, not zero error. Intersections of conditions may be even smaller.

How does correcting one B error affect its rate? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 31 • WORKED EVIDENCE

1. Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.
2. Accuracy gap = 90% − 50% = 40 percentage points.
3. Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = 80%.
4. Inspect representation and image quality; report counts and uncertainty before repairing.

STEP 1 Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.

STEP 2 Accuracy gap = 90% – 50% = 40 percentage points.

STEP 3 Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = **80%**.

STEP 4 Inspect representation and image quality; report counts and uncertainty before repairing.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Subgroup Performance

The facts we will use

AI ACADEMY / AI-4/31

Subgroup Performance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Sun rate

Sun error=2/20=10%.



2. Shade rate

Shade error=3/5=60%.



3. Pooled rate

Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

CHANGE ONE THING

Shade gains5 new cases with no errors.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a pooled score treated as proof for every group

> A gap is an observed difference,not an inherent group trait or a cause.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A good overall result guarantees acceptable performance for every **subgroup**.

Explain evidence and next-check limits

- Equal error rates alone do not establish fairness, and unequal rates alone do not identify discrimination or a cause. Groups may differ in base rates, label quality or conditions.

  Use results to ask better questions and plan review rather than label a group as inherently difficult.

> **Why does equal accuracy not settle whether use is fair?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A good overall result guarantees acceptable performance for every subgroup. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise a planned subgroup audit with less help

> Distinguish predeclared slices from exploratory searches.

Try supported sensitivity W2

Use workbook W2 for the connected example “**Subgroup** evaluation”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Retain definitions,error types,label quality and sampling

Choose group definitions for a reason connected to use and possible harm. Retain error types, reference-label quality and sampling method. Avoid searching many groupings merely to find a dramatic or reassuring number.

Report exploratory checks as exploratory.

Explain the denominator behind the result

- Use the worked case for **Subgroup** evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain why the pictured total needs disaggregation

- Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent X/Ycounts

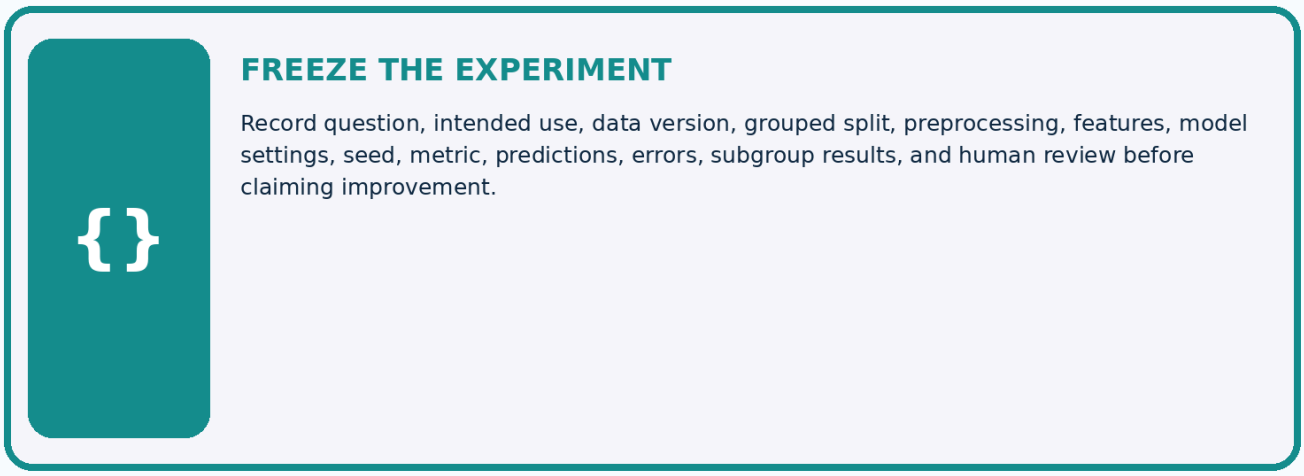
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + **denominator**s

PREDICTIONS + errors + **subgroup** slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Check sensitivity

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete lighting,type and camera Slice Audit

> All records needed for the activity's promised subgroup metrics are supplied.

Complete25-case source and prediction table

Case ID · Light · Toy type · Camera · Reference · Frozen prediction

> **Case ID:** A01
> **Light:** bright
> **Toy type:** Alpha
> **Camera:** A
> **Reference:** spotted
> **Frozen prediction:** spotted

> **Case ID:** A02
> **Light:** bright
> **Toy type:** Alpha
> **Camera:** A
> **Reference:** spotted
> **Frozen prediction:** spotted

Case ID: A03

Light: bright

Toy type: Alpha

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A04

Light: bright

Toy type: Alpha

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A05

Light: bright

Toy type: Alpha

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A06

Light: dim

Toy type: Beta

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A07

Light: dim

Toy type: Beta

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A08

Light: dim

Toy type: Beta

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A09

Light: dim

Toy type: Beta

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A10

Light: dim

Toy type: Beta

Camera: A

Reference: spotted

Frozen prediction: spotted

Case ID: A11

Light: dim

Toy type: Alpha

Camera: A

Reference: healthy

Frozen prediction: spotted

Case ID: A12

Light: dim

Toy type: Alpha

Camera: A

Reference: healthy

Frozen prediction: spotted

Case ID: A13

Light: bright

Toy type: Alpha

Camera: A

Reference: healthy

Frozen prediction: healthy

Case ID: A14

Light: bright

Toy type: Alpha

Camera: A

Reference: healthy

Frozen prediction: healthy

Case ID: A15

Light: bright

Toy type: Alpha

Camera: A

Reference: healthy

Frozen prediction: healthy

Case ID: A16
Light: bright
Toy type: Beta
Camera: A
Reference: healthy
Frozen prediction: healthy

Case ID: A17
Light: bright
Toy type: Beta
Camera: A
Reference: healthy
Frozen prediction: healthy

Case ID: A18
Light: dim
Toy type: Beta
Camera: A
Reference: healthy
Frozen prediction: healthy

Case ID: A19
Light: dim
Toy type: Beta
Camera: A
Reference: healthy
Frozen prediction: healthy

Case ID: A20

Light: dim

Toy type: Beta

Camera: A

Reference: healthy

Frozen prediction: healthy

Case ID: B01

Light: dim

Toy type: Beta

Camera: B

Reference: spotted

Frozen prediction: healthy

Case ID: B02

Light: dim

Toy type: Beta

Camera: B

Reference: spotted

Frozen prediction: healthy

Case ID: B03

Light: dim

Toy type: Beta

Camera: B

Reference: spotted

Frozen prediction: spotted

Case ID: B04

> **Light:** dim
>
> **Toy type:** Beta
>
> **Camera:** B
>
> **Reference:** healthy
>
> **Frozen prediction:** spotted

> **Case ID:** B05
>
> **Light:** dim
>
> **Toy type:** Beta
>
> **Camera:** B
>
> **Reference:** healthy
>
> **Frozen prediction:** healthy

Definitions,positive class and source-count contract

Slice Audit uses the complete 25-case synthetic record above, one case per fictional plant, and spotted as the toy positive class. Audit lightingbright/dim, typeAlpha/Beta, cameraA/B/C. Camerachas no evaluated cases. Within each field the declared observed groups are disjoint; the same case appears again across different fields, so do not add lighting+type+camera totals as if they were new independent observations.

These saved predictions are supplied, not a newly executed classifier or actual species study.

Count,compare and prioritize a justified investigation

1. Compute error rates and spotted recall for every declared lighting, type and camera slice from the full case table. Identify all numerators and denominators before comparing percentages, and mark missing-camera coverage. Choose one gap to investigate using representation, label quality and collection context; do not name a cause from this table alone.

   Compare your counts with the optional complete saved-output program and fold-out key. Keep source IDs so exploratory intersections do not create extra independent cases. Reserve the new X/Ytask for an uncoached first attempt.

▼ Check all slice metrics after trying

Slice	Error count / total	Found spotted / reference spotted	Interpretation
Bright	0 /10	5 /5	Zero observed errors on10 cases does not establish perfect future behavior.
Dim	5 /15	6 /8	Error33.3%, recall 75%; lighting co-varies with other fields.
Alpha	2 /10	5 /5	Error20%, recall 100%on these supplied records.
Beta	3 /15	6 /8	Error20%, recall 75%; equal error rates do not imply equal recalls.
CameraA	2 /20	10 /10	Error10%, recall 100%.
CameraB	3 /5	1 /3	Error60%, recall 33.3%; one result strongly changes these small denominators.
CameraC	0 /0	0 /0	No observed error rate or recall; do not call missing coverage perfect.
Pooled once	5 /25	11 /13	Error20%, recall≈84.6%; retain all slice tables.

Normal,small-sample,missing-coverage and overlap probes

Complete subgroup probe	Expected bounded check
Normal disjoint cameraA/Bpartition	5/25 pooled error, not(10%+60%)/2=35%.
Boundary cameraBhas5 cases	One changed outcome shifts error rate by20 percentage points; retain source count.
Missing cameraCexamples	Report no evidence, not zero measured error. Unknown metadata also needs its own documented status.
Overlapping exploratory slices across fields	Do not double-count pooled cases or turn a found difference into a predeclared causal finding.

Optional complete slice audit program

```
# Complete synthetic saved-output realization of camera A:2/20 and B:3/5 errors.
records = []
for i in range(1, 21):
    records.append({'id': f'A{i:02}',
                    'light': 'bright' if i <= 5 or 13 <= i <= 17 else 'dim',
                    'type': 'Alpha' if i <= 5 or 11 <= i <= 15 else 'Beta',
                    'camera': 'A', 'actual': 'spotted' if i <= 10 else 'healthy',
                    'prediction': 'spotted' if i <= 12 else 'healthy'})
for i in range(1, 6):
    records.append({'id': f'B{i:02}', 'light': 'dim', 'type': 'Beta', 'camera': 'B',
                    'actual': 'spotted' if i <= 3 else 'healthy',
                    'prediction': 'spotted' if i in (3, 4) else 'healthy'})


def summarize(rows):
    errors = sum(r['actual'] != r['prediction'] for r in rows)
    positives = sum(r['actual'] == 'spotted' for r in rows)
    found = sum(r['actual'] == r['prediction'] == 'spotted' for r in rows)
    return {'errors': errors, 'count': len(rows),
            'error_rate': errors / len(rows) if rows else None,
            'positive_found': found, 'positive_count': positives,
            'recall': found / positives if positives else None}
```

```
for field, groups in [('light', ['bright', 'dim']),
                      ('type', ['Alpha', 'Beta']), ('camera', ['A', 'B', 'C'])]:
    for group in groups:
        print(field, group, summarize([r for r in records if r[field] == group]))
print('pooled', summarize(records))
# Each case is one distinct fictional plant; groups partition within each field.
# Across fields the same cases recur, so do not sum lighting+type+camera counts.
# Inputs are complete resolved reference/prediction records, not model training.
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted

> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

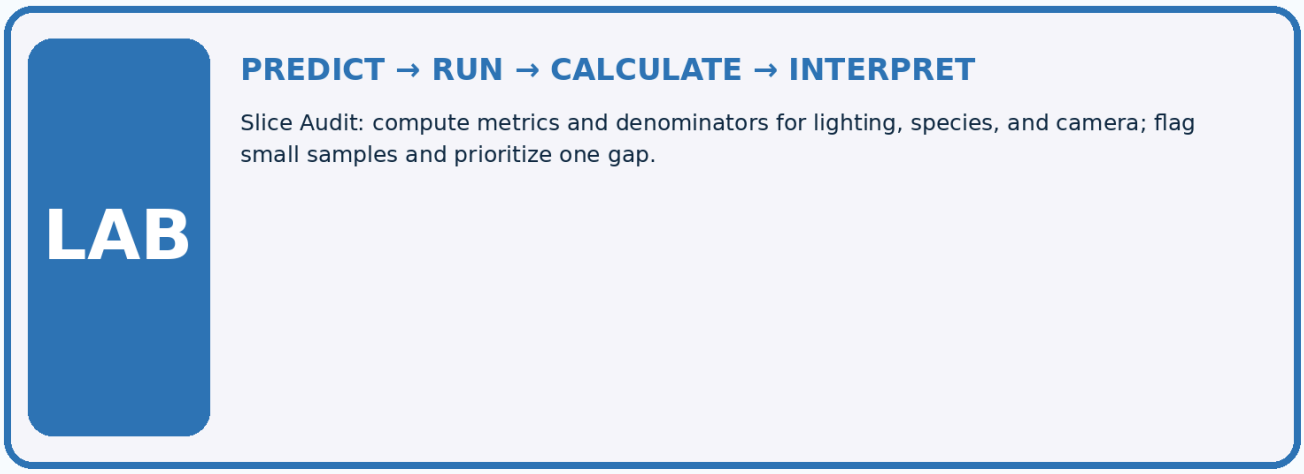
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Slice Audit: compute metrics and **denominator**s for lighting, species, and camera; flag small samples and prioritize one gap.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.
</details>

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare two fresh subgroup error counts

> State the weighted total and one evidence limit.

Complete independent X/Ycounts and scope

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

Treat the selected X/Ygroups as the supplied disjoint 14-case comparison with the same error definition. Do not use the main 25-case camera table or invent a causal group difference. Report each count, pooled denominator and the four-case Ysample limitation. No per-class labels are given, so recall is not computable from these aggregate error counts alone.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep project release discussion tied to subgroup evidence

> An overall pass should not erase unmeasured or weakly supported conditions.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

Keep subgroup definitions, source/sample counts, positive-class counts, reference quality, metric versions and all slice results with the project. Label exploratory checks honestly. A proposed classroom minimum threshold is a development policy requiring justification and adequate evidence, not a validated certification or guarantee.

Missing or very small groups need explicit limitations and an evidence-collection plan rather than a hidden favorable total.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

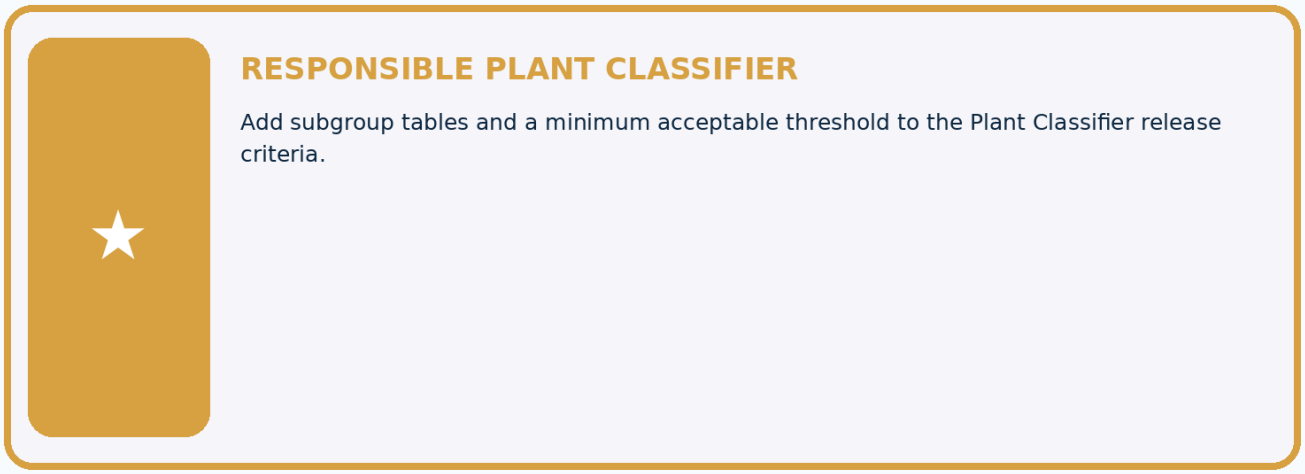
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with the slice definition,count and unanswered cause question

> Explain what the total hides and what new evidence should be collected.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Calculate **subgroup** results with sample sizes

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
subgroup	meaningful slice of evaluation cases	
performance gap	difference between subgroup metrics	
denominator	number of cases behind a rate	
intersection	cases belonging to multiple slices	

Closer-explanation terms

Word	Meaning
Subgroup	A defined subset of the evaluated examples
Disaggregation	Reporting results separately for defined groups
Coverage	Which groups and conditions are represented

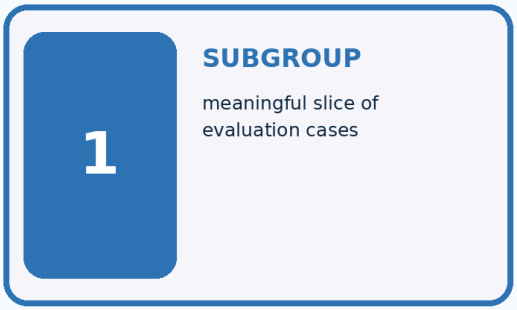
Word	Meaning
Disparity	An observed difference between group results

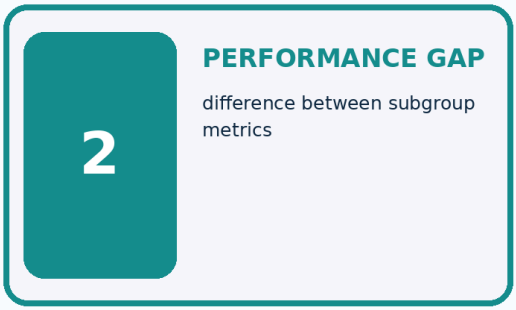
▼ Earlier wording, alternatives and picture transcripts

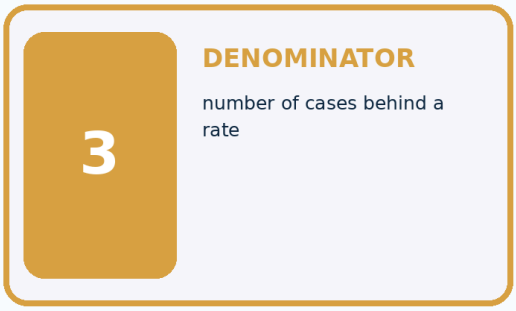
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A good overall result guarantees acceptable performance for every **subgroup**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **subgroup**, **performance gap**, **denominator**, **intersection**.
- CALCULATE Reproduce the worked **evidence** with units and **denominator**.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A good overall result guarantees acceptable performance for every **subgroup**.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

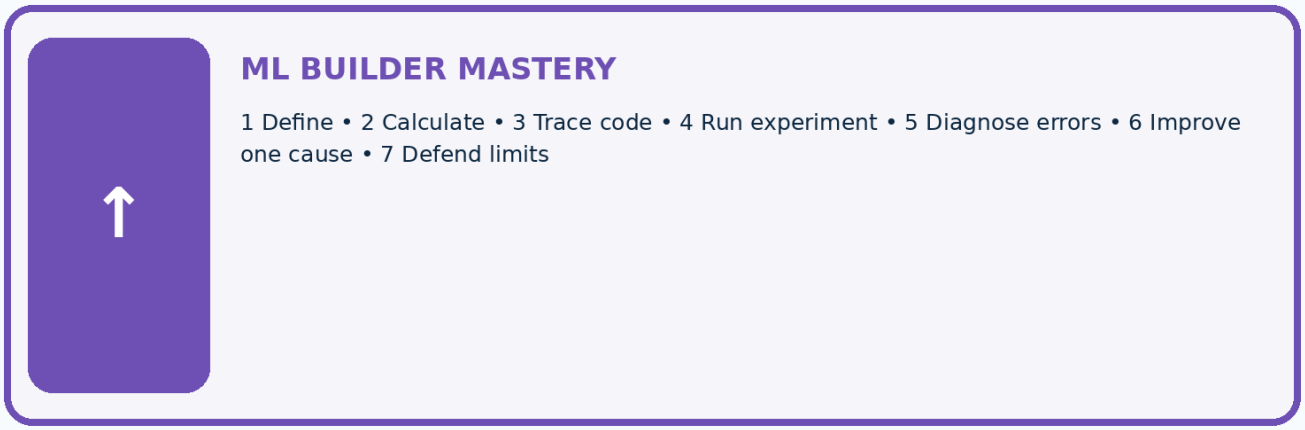
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh X1/10andY2/4error-rate task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

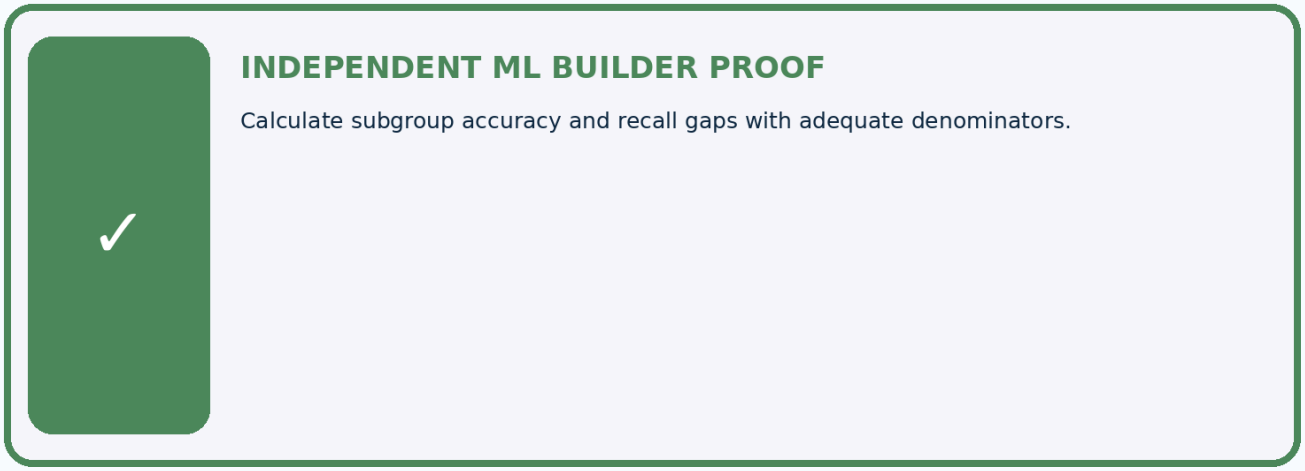
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Calculate **subgroup** accuracy and recall gaps with adequate **denominator**s.

NEXT BRIDGE Carry the experiment **evidence** into: Proxy Features and Fairness.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A has18/20 correct; B has6/10.

Compute **subgroup** results and **evidence** limits.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

Evidence or stage	Value or result
Group A	18/20
Group B	6/10

▼ Earlier wording, alternatives and picture transcripts

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Report subgroup denominators with rates

A has18/20 correct; B has6/10.

Compute **subgroup** results and **evidence** limits.

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Calculate subgroup results with sample sizes	Explain the deciding step, show a trace or test, and state one limit.
Distinguish missing evidence from equal performance	Explain the deciding step, show a trace or test, and state one limit.
Report differences without assigning causes from a table	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

- Which labelled step matches this detail? Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

- 1. Pooled rate

- 2. Sun rate

- 3. Shade rate

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Pooled rate Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Read the picture in order.

Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Shade gains5 new cases with no errors. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Repair and keep the record

Review Improve Generalization (ai-4/30). Goal: Plan one **evidence**-led change and test it on source-separated validation cases. Short practice: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work
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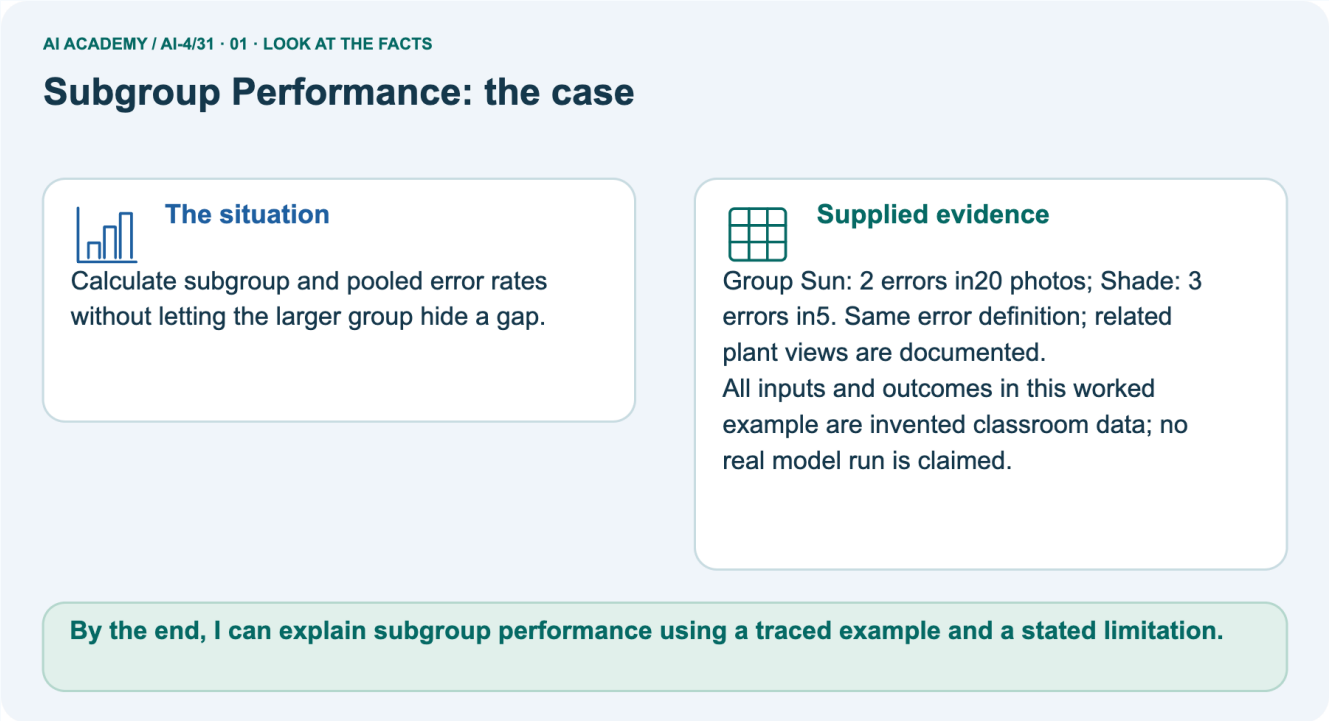
<details>
<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Subgroup Performance: the case



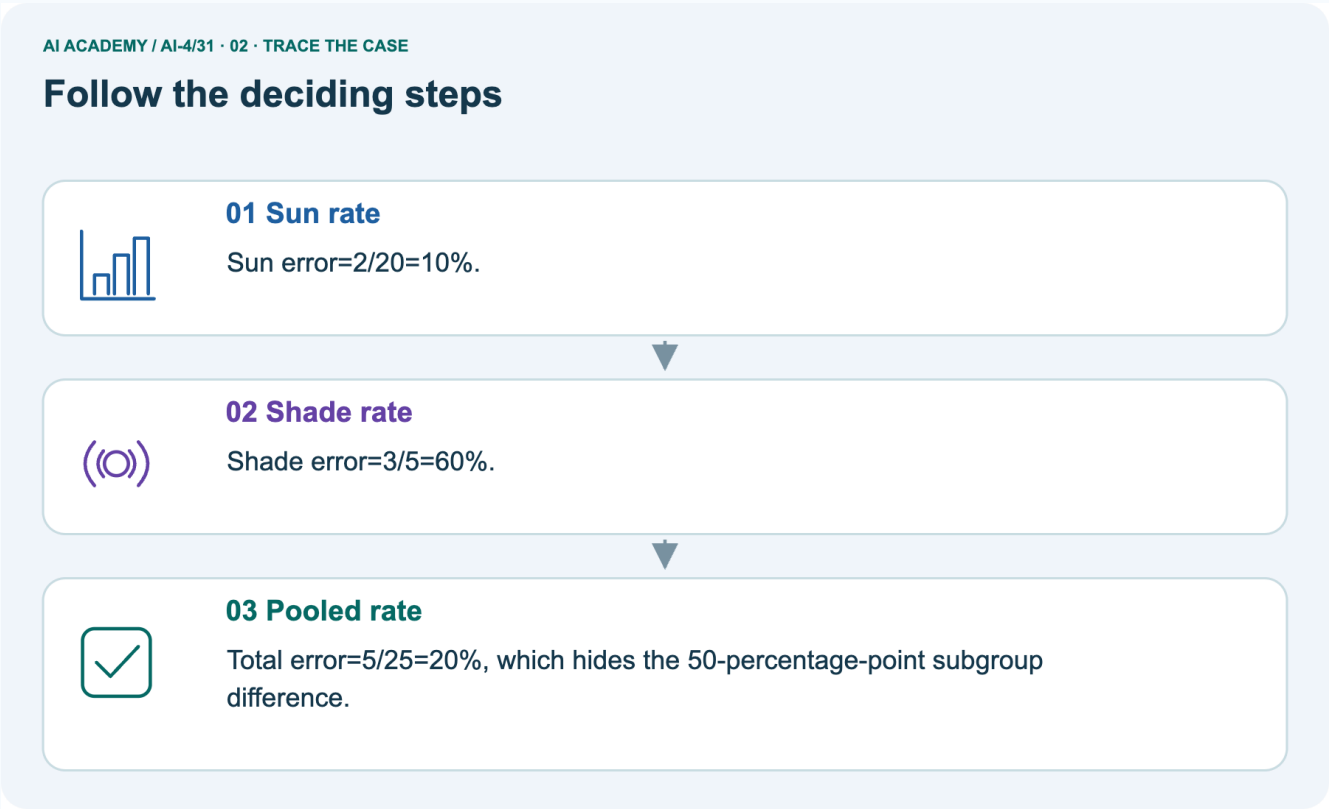
Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap. Supplied facts: Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Sun rate: Sun error=2/20=10%. → Shade rate: Shade error=3/5=**60%**. → Pooled rate: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

Point to each numbered stage and explain which recorded **input** it uses.

03. Subgroup error rates

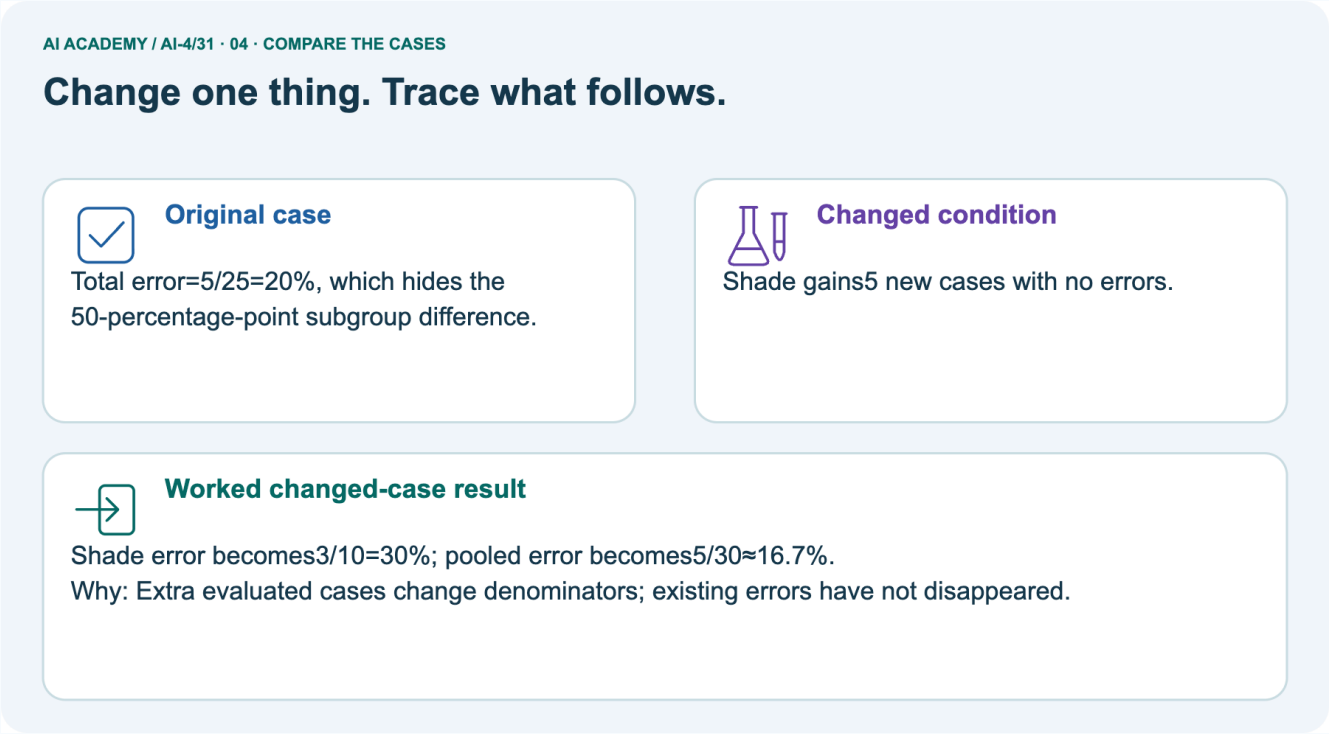


Follow the labelled picture with the matching explanation. Open the vector diagram

Subgroup error rates. Units: percent. Sun (n20): 10. Shade (n5): 60. pooled (n25): 20.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



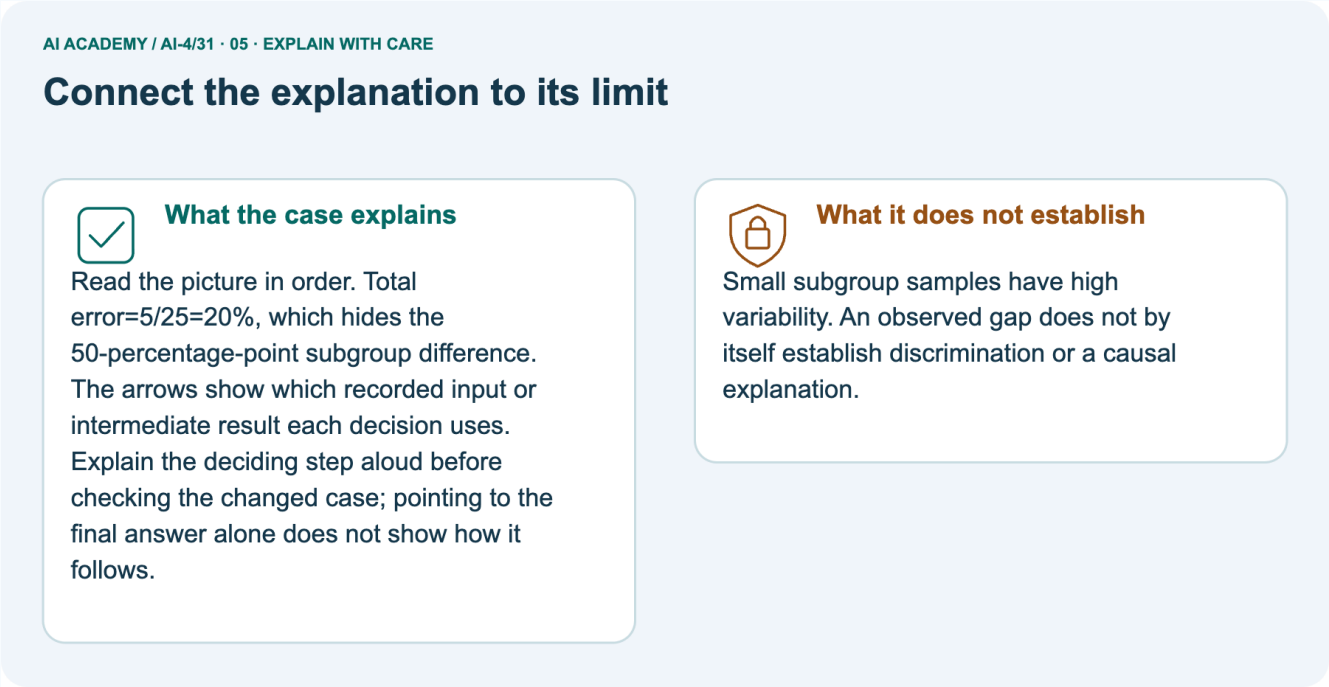
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Changed condition: Shade gains5 new cases with no errors.

Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

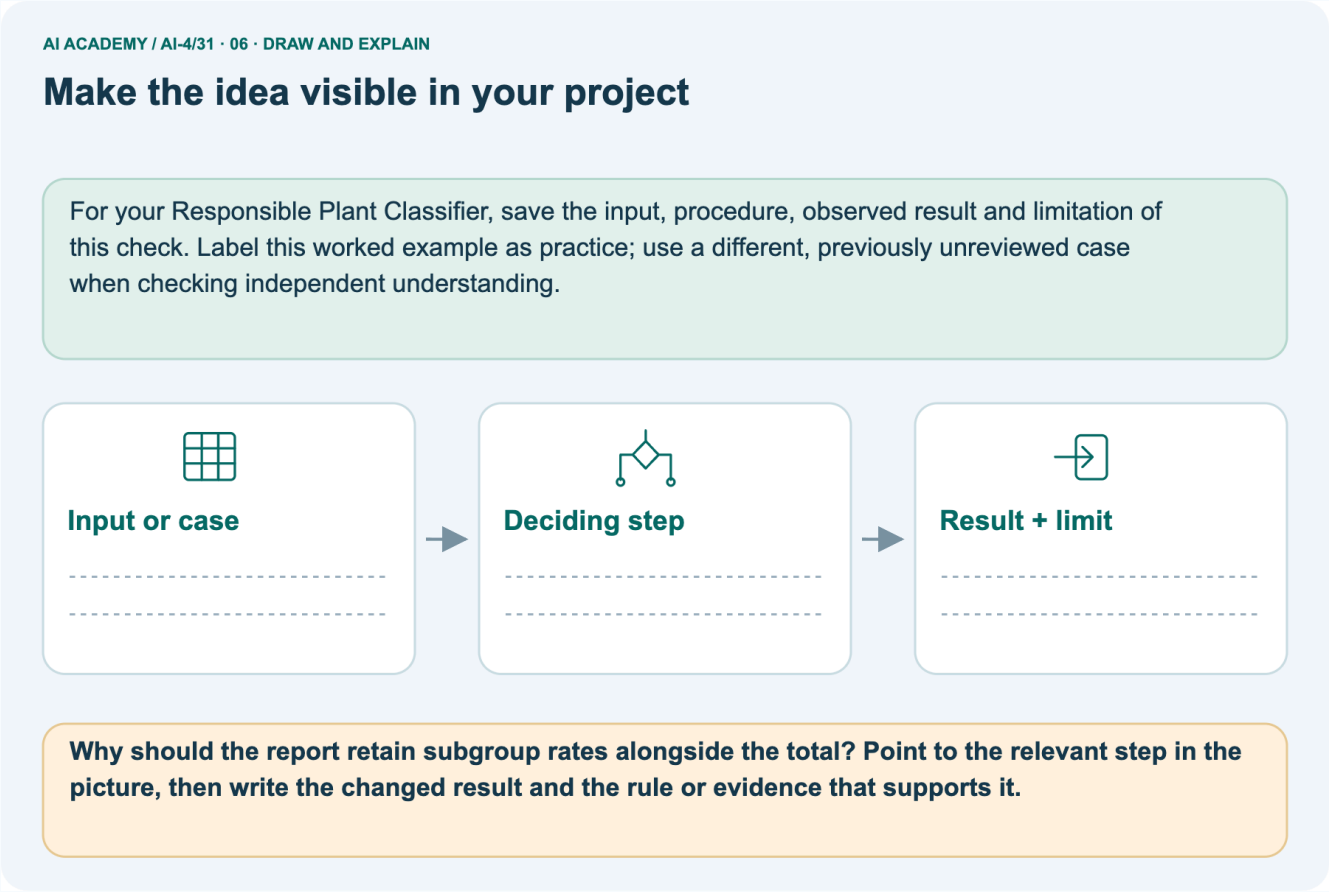
Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Small **subgroup** samples have high variability.

An observed gap does not by itself establish discrimination or a causal explanation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Subgroup Performance: the case

02. Follow the deciding steps

03. Subgroup error rates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.